A320 FAL Digital Twin

See where final assembly loses time, and test the fix before it reaches the shop floor.

Concept one-pager · October 2026
Prepared by Miguel Figueroa
SYNTHETIC DATA

THE IDEA

An interactive 3D twin of an A320 Family final assembly line, from section delivery (S50) to flight line. It replays the line at takt, traces the **critical path**, ranks **bottlenecks** and quantifies **rework**, travelled work and gliders. Every improvement lever, rebalancing move or disruption is re-simulated, so decisions come with a measured effect on **rate, lead time and delivery dates**, not a guess.

WHAT THE MODEL SHOWS · ILLUSTRATIVE

PLAN RATE	LEAD TIME	FIRST-PASS YIELD
10.5 /mo	**≈ 25** d	**≈ 33** %
STATIONS	**LINE POWER**	**IMPACT RUNS**
8 + 3 feeders	**1.9–2.7** MW	**3 × 200** d



Live line with the aircraft inspector (status, ETA vs. commitment, station-by-station journey) and the searchable fleet list. Station tags carry live sensor values.

Capabilities

Live line simulation
Discrete-event model of 8 stations, bays, engine supply and gliders; takt 2.0 / 1.8 / 1.6 d.

Critical path & balance
CPM Gantt with float on feeders; drag-and-drop line balancing (yamazumi) against takt.

Inspect any asset
Click an aircraft or AGV: status, progress, customer, ETA, travelled work, predicted escapes.

Fleet & orders
Aircraft, bays and AGVs with search; orders by customer and the next launch slots.

Alerts & predictive maintenance
Bottleneck, engine shortage, late MSNs, robot vibration and paint-booth humidity, ranked by severity.

Twin assistant
For each alert: situation, cost of doing nothing, ranked options with effect and cost, root cause. Apply changes the simulation.

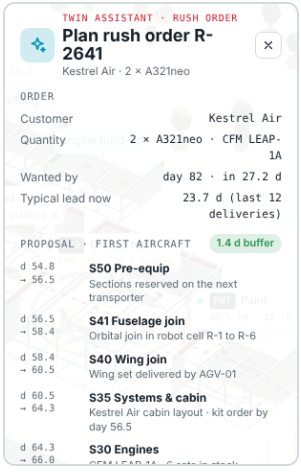
Rush-order planning
Slot proposal with day-by-day timeline, buffer vs. wanted date, kit-order deadline, what moves, what was ruled out.

Power load balancing
Load per station, 12 h peak forecast vs. contract peak; stagger cure ovens or shave with a site battery.

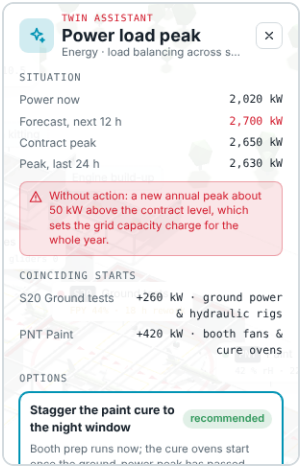
How a decision flows

1. **Sense.** Twin feeds (station progress, robot spindles, booth climate, supplier ETAs, power) update every bay and aircraft.
2. **Detect.** Rules flag bottlenecks, late aircraft and sensor trends before they hit the line, e.g. robot R-3 reaching its vibration limit in ≈ 9 d.
3. **Decide.** The assistant compares options with simulated gains: second S40 jig, third S20 shift, floating team, kit gate, edge-AI hole inspection.
4. **Prove.** The impact report re-runs plan vs. change for 200 days × 3 seeds, under the same disruptions, and logs every decision for the shift handover.

Built as one self-contained HTML page: three.js isometric scene, in-browser simulation and CPM, three view layers (physical, digital twin, metaverse) plus a thermal heat view. The assistant is rule-based on the twin's model, not a language model.



Rush-order planner



Power-peak assistant

All layouts, cycle times, rates, defect rates, costs, customers and power figures are synthetic, invented for this concept. They do not represent any manufacturer's data.